LOCAL FINITENESS OF TYPE IS PERMANENT ON THE SOURCE OF A COMPOSITION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof span cites one same-chapter lemma and one cross-chapter lemma of Morphisms of Spaces; both are delivered as navigation stubs. No bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Permanence of representability by algebraic spaces on the source of a composition. Same chapter, Lemma lemma-permanence-representable, invoked at proof lines 5766-5767. Shows that f is representable by algebraic spaces, so that it may be tested after base change to a scheme.
- Permanence of local finiteness of type on the source of a composition of algebraic spaces. Morphisms of Spaces, Lemma spaces-morphisms-lemma-permanence-finite-type, invoked at proof lines 5769-5771. Applied to the base-changed morphisms of algebraic spaces to conclude.

Source reference. Original lemma 0AQ0: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4076–4076. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0462: sources/stacks-chow/tags.txt, source lines 5457–5457. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ and $g : Y \to Z$ be morphisms of formal algebraic spaces over S. If $g \circ f : X \to Z$ is locally of finite type, then $f : X \to Y$ is locally of finite type.*

Proof. By Lemma 0AQ0 we see that f is representable by algebraic spaces. Let T be a scheme and let $T \to Z$ be a morphism. Then we can apply Morphisms of Spaces, Lemma 0462 to the morphisms $T \times_Z X \to T \times_Z Y \to T$ of algebraic spaces to conclude. □

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